

Pritam

Portfolio: prtmxio

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Education

Deenbandhu Chhotu Ram University of Science and Technology

2023 – Present

B.Tech, Electronics & Communication Engineering; CGPA: 8.49/10

Sonapat, Haryana

CBSE Class XII

2022

Mathematics – 96.2%

New Delhi

Experience

Embedded Systems Intern

Mar 2026 – Present

Glazia

Remote

- Built **Glazia Home Secure**, an ESP-IDF IoT home security system: hub firmware (ESP32-S3) driving an LVGL touch display, fingerprint auth, camera capture, and environmental sensors.
- Fixed a camera DMA bottleneck by retuning OV5640 XCLK to **16MHz**, sustaining **10+ fps** MJPEG capture.
- Built an encrypted **ESP-NOW** sensor mesh with BLE provisioning and a secure multi-party pairing protocol.
- Designed dual-factor device authentication requiring MAC address and a server-issued secret to match.
- Prevented spoofed hub identity across all device routes via strict credential validation.

Embedded Application Developer (Intern)

Jul 2025 – Dec 2025

PCB CUPID

Bangalore, Remote

- Co-built **Circuit Spell** (spell.pcbcupid.com), an AI agent generating full PCB designs from natural language using 50+ proprietary Glyph modules.
- Prototyped EDA subsystems for constraint-based component placement and auto-routing.
- Developed multi-sensor IoT firmware in C/C++ on Glyph boards for industrial control.

Robotics Engineering Team Member and Mentor

Oct 2023 – Present

THINKBOTS Robotics Club, DCRUST

Murthal, Haryana

- Designed autonomous line-following robots using Arduino with PID control, achieving 95% path accuracy on competition tracks.
- Mentored 15+ junior developers in embedded systems, PID control, and algorithm design.

Projects

meshflow | Python, ROS 2, Gazebo, Onshape API, URDF/Xacro | 🐙

June 2026

- Built a CLI tool converting Onshape CAD assemblies into simulation-ready ROS 2 packages (URDF, xacro, Gazebo plugins, launch files) via the Onshape REST API.
- Designed **KinematicDAG**, a geometry-only classifier inferring robot type and sensor kind directly from URDF joint structure, with zero manual naming conventions required.
- Auto-generates Gazebo/ros2_control plugins with wheel geometry measured via **trimesh** for physical accuracy.

synapse | Transformers, Diffusers, HNSW | 🐙

Mar 2026

- Local multimodal pipeline: **SmolVLM-500M** (caption) → **CLIP** (retrieval) → **SD-Turbo** (generation) within just **4 GB VRAM** budget.
- CLIP retrieval via **HNSW ANN indexing**, achieving **~1ms** query latency over 5,000+ images.
- End-to-end latency: **~6s** (text) / **~12s** (image) on RTX 2050 via co-loaded models.

xio_sat | ESP-IDF, ForgeFPGA, ROS 2, MicroPython | 🐙

Feb 2025

- Fault-tolerant CubeSat guardian node: FPGA watchdog across a 4-node stack, guaranteeing hard reboot within just **1.5s** of fault, with reset cause persisted to flash.
- ADCS telemetry pipeline: IMU → RP2040 filter → ESP32-C6 → MQTT → ROS 2, validated via fault injection.
- ROS 2 ↔ MQTT bridge multiplexing **3 telemetry streams at 20Hz** with bidirectional command relay.

Technical Skills

Programming Languages: C/C++, Python, Java, Verilog

Embedded & Robotics: ESP-IDF, ROS 2, Arduino, ForgeFPGA, MQTT, PID Control, Firmware Development

AI / ML: PyTorch, HuggingFace, Transformers, Diffusers, CLIP, OpenCV, Scikit-learn, CNN

Tools: Git, Linux, KiCad, VS Code, (neo)Vim, Gazebo, URDF, Onshape API

Achievements

Smart India Hackathon | *Team Leader & Software Developer*

2024

- Built a real-time AI face recognition system for missing persons identification, deployed at Simhastha Ujjain Mela in collaboration with the Police Department.